

Outcome of resective surgery for intractable partial epilepsy guided by subdural electrode arrays

Jennum P, Dhuna A, Davies K, Fiol M, Maxwell R. Outcome of resective surgery for intractable partial epilepsy guided by subdural electrode arrays.

Acta Neurol Scand 1993; 87: 434–437. © Munksgaard 1993.

The aim of this paper was to evaluate the outcome and the factors predictive for a good prognosis of resective surgery for intractable partial epilepsy guided by subdural electrode arrays (SEA's) and operative electrocorticography. Sixty-four patients, aged 8–52 years, were evaluated with chronic SEAs in order to record interictal and ictal activity and delineate speech and motor areas by functionally mapping. Resection were individualized to each patient's SEA recorded electrocorticogram and operative electrocorticogram and functional mapping results (tailored resection). The follow-up time was a minimum of one year. Good seizure outcome was defined as seizure free from complex partial and secondary generalized seizures. After one year 70% of the patients with a temporal ictal focus was seizure free compared to 55% of the patients with an extratemporal focus. Complete resection of interictal or ictal fields as mapped with SEAs, gave better prognosis than partial resection. Patients with no postresection spikes had a better prognosis than patients with residual postresection spikes evaluated with operative electrocorticography. Sex, age, duration of epilepsy prior to surgery, extent of temporal lobe resection and structural abnormalities determined by MRI were not associated with a favorable seizure outcome after surgery. We conclude that complete resection of the interictal and ictal field mapped with SEA's and absence of postresection spikes on operative electrocorticography are associated with an excellent seizure outcome.

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Key words: invasive electrophysiology; subdural electrodes; electrocorticography; partial epilepsy; prognosis; surgery

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Accepted for publication December 7, 1992

Recordings from the scalp surface are not always able to precisely localize the site of ictal onset of partial seizures secondary to artifacts, attenuation or smearing of the surface EEG by the dura, skull and scalp. Implanted electrodes are sometimes necessary to localize site of ictal onset during the presurgical evaluation of patients with medically intractable seizures. Since Penfield & Jasper (1), Lilly et al. (2) and Abraham & Ajmone-Marsan's (3) description of the clinical and experimental use of subdural electrodes, a number of laboratories have utilized this technique, implantation of strips or arrays of electrodes in various portion of the epidural or subdural space (4). Localization of motor, sensory and language-related cortex can be determined by electrical stimulation of the subdural electrodes. By this information regarding the epileptogenic focus and the relation to important functional brain areas are obtained allowing tailored resections in order to po-

tentially improve the treatment results and reduce the risk of neurological deficits (5).

The aim of this paper was to evaluate the outcome and the factors predictive for a good prognosis of resective surgery for intractable partial epilepsy guided by subdural electrode arrays (SEA's) and operative electrocorticography.

Methods

Patients

During January 1983–1990, 94 patients with medically intractable epilepsy underwent a "tailored" temporal or cortical resection. In all these cases the SEA's were placed under the supervision of one investigator (R.M.). Moreover, an operative electrocorticography was performed as well. Of these 64 had a follow-up of one year.

Nineteen were women and 45 were men (n = 64).

The median age was 24 years (range: 8–52 years). Twenty patients were 18 years of age or younger. The median duration from the first seizure to surgery was 13 years (range 1–38 years). Four patients had undergone a previous temporal lobectomy.

Non-invasive EEG evaluation

All patients had a preoperative evaluation with scalp video-EEG recordings for a median 11 days (range 2–41 days) prior to the implantation of subdural electrodes. A 32-channel video-EEG was used (Telefactor Corporation, Philadelphia, USA). The median number of seizures recorded in this period was 9 (range 2–53). Standard scalp electrodes were placed according to the international 10–20 system, and with additional bilateral sphenoidal wire electrodes.

Subdural EEG evaluation

The scalp sphenoidal video-EEG data, intracarotid amytal test (6), neuropsychological data, MRI- and CT-scan were used to plan the placement of the SEA. Most patients had a 64 SEA's (platinum stainless steel contacts with 1 cm center to center separation) and one to four additional sentinels (1 row of 4 electrodes). The SEA's were placed in various locations to cover as completely as possible the expected area of epileptogenicity and surrounding functional tissue. Pre-implantation electrocorticography was obtained in many patients to assist in placement of the SEA's.

Video-electrocorticography recording with subdural electrodes were performed during decrease or discontinuation of anticonvulsive medication. A 64 channel video-EEG was used (Telefactor Corporation, Philadelphia, USA). The subdural electrodes were in place for median 16 days (range 3–32). During this period median 9 (range 1–34) seizures were recorded. Ictal onset was interpreted by an experienced electroencephalographer. The interictal maps were generated from daily interictal baseline recordings during wakefulness with photic stimulation and if possible during sleep.

Functional mapping studies

Hemispheric speech dominance and memory functions were determined by intracarotid injection of amobarbital (6). Fifty four patients had left sided speech dominance, 6 right sided, 2 bilateral speech dominance and was inconclusive in 2 patients. Functional mapping was performed by electrical cortical stimulation (7). Electrical stimulation (bipolar and referential) was performed using a Grass S-88 and a opto-electric isolator with a stimulation rate of

25 Hz biphasic pulses with a duration of 0.3 msec. The duration of the traces was 2–10 seconds with gradually increasing current from 1 to 5 mA. Language function (speech arrest) and motor activity were evaluated on each electrode site. The current intensity was adjusted to minimize afterdischarges. Stimulations were performed by several staff physicians and results recorded in the patients medical records. All of the medically records pertaining to stimulation were reviewed and tabulated specifically for this study on special forms.

Resections

Resections were individualized to each patients SEA and functional maps. Areas with ictal, interictal and operative epileptiform discharges were resected as completely as possible while avoiding any functional areas by at least 1 cm.

In 35 (55%) patients the resection was performed in the dominant side. Forty-four patients had a temporal lobe resection. In 20 patients an extratemporal resection was performed, including resection of the frontal lobe (13 patients), parietal lobe (4 patients) and occipital lobe (3 patients). Of the 20 patients with extratemporal resection a temporal lobe resection was performed in 7 patients in addition. The median extent of dominant hemispheric temporal resections from the temporal pole was 5.5 cm (range 4–9 cm), whereas the median extent for non-dominant hemispheric temporal resections were 6.5 cm (range 5–10 cm).

After resections postoperative electrocorticography was performed. Residual significant interictal spikes were resected if they did not extend into cortex with speech or motor activity mapped during the stimulation studies.

Statistics

All data are presented as mean (range) or percent (number of patients seizure free compared with the total number of patients in that particular group). Mann-Whitney, Wilcoxon Matched paired tests or chi-square test were used where appropriate. A stepwise multiple regression analysis was applied on dichotomous variables in order to evaluate the influence of different factors on surgical outcome. Statistical level of significance was $p < 0.05$.

Results

Clinical outcome

Good seizure outcome was defined as seizure free with absence from complex partial and secondary generalized seizures. To determine outcome, all of the patients charts were reviewed by the authors.

Table 1. Seizure activity before and after resection. Extratemporal and temporal foci. Number of seizures per month. Mean (range).

Seizure type	Extratemporal N=20			Temporal N=44			P (Extratemporal versus temporal)
	Presurgical	One year follow-up	P ¹	Presurgical	One year follow-up	P ¹	
Simple partial	28.5 (0–120)	12.3 (0–65)	n.s.	12.0 (0–105)	2.4 (0–30)	≤0.0002	n.s. ²
Complex partial	45.0 (0–300)	2.3 (0–10)	<0.02	13.3 (0–90)	3.4 (0–100)	<0.00001	n.s. ²
Secondary generalized	13.4 (0–150)	1.9 (0–20)	<0.01	14.7 (0–450)	0.3 (0–5)	≤0.0002	n.s. ²
Overall	86.9 (3–301)	16.4 (0–70)	<0.02	40.3 (3–456)	6.0 (0–100)	<0.00001	n.s. ²
Seizure free (complex partial and secondary generalized seizures)		55%			70%		n.s. ³

¹ Wilcoxon matched paired test.² Mann-Whitney test between presurgical values: extratemporal versus temporal foci.³ Chi-square test.

Seizures occurring within 3 month of surgery were not used.

The presurgical seizure frequency per month tended to be higher in extratemporal foci (defined as extratemporal with or without a temporal focus), but was not statistical significant. 70% of the patients with a temporal focus and 55% of the patients with an extratemporal focus were seizure free one year after surgery (n.s.) (Table 1).

The following MRI-findings were observed: tumor (14 patients), localized temporal/hippocampal atrophy (7 patients), hamartoma (2 patients), former stroke (2 patients), arteriovenous malformation (1 patient), the rest (38 patients) were normal.

The median presurgical verbal intelligence quotient (IQ) was 92 (64–136), perceptive IQ was 95 (51–121) and full scale IQ was 94 (range 64–130).

No relation was found between age, duration of epilepsy prior to surgery, sex, MRI abnormality (residual temporal sclerosis, tumor, arteriovenous malformation etc.) and seizure outcome. Localization of ictal focus, whether ictal or interictal focus were completely resected and postresection spikes all showed significant relation to good seizure outcome (Table 2). In a stepwise multivariate analysis, the

factors most strongly associated with good seizure outcome, taking other factors into consideration, were resection of ictal field ($p < 0.01$) and absence of postresection spikes ($p < 0.05$).

In patients with no postresection spikes and in whom all ictal field were resected ($N = 45$), 82% were seizure free. If ictal mapped field were not fully resected but with no remaining postresection spikes 50% were seizure free. If all ictal field were resected, but with the presence of postresection spikes, then 55% were seizure free. In one patient the only parts of ictal field were resected and in which postresection spikes were present were not seizure free.

Complications

One case developed a scalp infection. None of the patients presented aphasia at the neurophysiological evaluation 6 month after surgery.

Discussion

Our study suggest that the factors of importance for a good seizure outcome are a complete resection of the ictal field and absence of postresection spikes.

Table 2. The effect of different factors influencing the prognosis one year after resection. Patients with implanted subdural electrodes. Number of patients (percent) of patients seizure free after resection in each of the groups. Uni- and multivariate analysis. In total 42 patients of 64 (66%) were seizure free from complex partial and secondary generalized seizures.

Variable	Percent (numbers/total in each group) seizurefree	P Chi-square tests
Age: ≤18 versus <18 years	60 (12/20) versus 68 (30/44)	n.s.
Sex: female versus male	63 (12/19) versus 67 (30/45)	n.s.
Duration: ≤10 versus >10 years	63 (15/24) versus 68 (27/40)	n.s.
MRI: structural lesion (tumor, arteriovenous malformation or hamartoma) versus normal or localized atrophy*		
Extend of temporal resection; >5 cm versus ≤5 cm	64 (30/44) versus 60 (12/20)	n.s.
Temporal versus extratemporal focus	70 (31/44) versus 55 (11/20)	<0.05
Ictal focus completely resected: yes/no	70 (35/50) versus 43 (6/14)	<0.05
Interictal focus completely resected: yes/no	79 (33/42) versus 41 (9/22)	<0.02
Presence of postresection spikes in the operative electrocorticography: yes/no	82 (31/38) versus 42 (11/26)	<0.005

* 7 patients showed localized temporal atrophy. Four (57%) of these were seizure free.

The best seizure outcome in patients with a temporal focus are comparable to the results from patients with epilepsy surgery with standard temporal resection based upon surface EEG data and congruent diagnostic results (8–10). Patients with an extratemporal focus showed a slightly poorer prognosis. This difference was not significant, but may be due to a type 2 error. The results are similar to other results (11–14). The indications using SEAs in the patients with a temporal focus was presence of speech dominance or a possible extratemporal focus. Extratemporal foci may show a poorer prognosis due to a potential larger focus or because the focus coincided with functional eloquent cortex limiting a complete resection.

Similar to other studies we found that age, sex, duration of epilepsy, extent of temporal resection or presence of localized abnormality at the MRI scan did not influence the seizure outcome (12–15).

Scalp EEG recordings, particularly if the ictal onset is neocortical and/or extratemporal may not be accurate in order to localize the ictal/interictal regions. This is due to artifacts from movements and muscle, and the attenuation and distortion of the EEG by the dura, skull and scalp. Electrocorticography appears to provide a reliable intraoperative mapping of the extent of the interictal epileptogenic region in patients with complex partial seizure of temporal origin submitted for surgical therapy (8–10). Intraoperative cortical stimulation can also be used to map eloquent cortex but requires the patient to be awake during the surgery (8, 11). However, not all patients tolerate this procedure, especially children. SEAs have the advantage that ictal onset can be determined, and functional studies can be performed with minimal discomfort and can be repeated for validation. The disadvantages are the need for two surgeries with the additional operative risk from implantation of SEAs.

Wyler et al. (12) found the subdural technique effective, well tolerated and with a good clinical outcome in children and adults. They found a high infection rate (22%) which declined (7%) after changing the cabling technique and using prophylactic antibiotics.

Our infection rate was 2%. Prophylactic antibiotics were used in all patients. The most often encounter symptoms were headache and nausea related to increased intracranial pressure due to epigrid fluid and a cerebral oedema.

In summary, we have found SEAs invaluable in both children and adults to delineate the epilepto-

genic interictal/ictal field from functional brain areas in patients submitted for epilepsy surgery and unable to have standard temporal resection or extratemporal resections. Complete resection of SEA mapped interictal and ictal activity and no remaining spikes on postoperative electrocorticography are associated with an excellent seizure outcome.

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